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Exposé for Media Project

Adaptive Matched Filter in Frequency Domain for Sound Source Separation

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Ilmenau, January , 2025

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1. Introduction

Sound source separation in complex acoustic environments, such as reverberant rooms, remains a significant challenge in audio signal processing. These environments often involve overlapping audio sources, making it difficult to accurately distinguish and isolate individual signals. This project focuses on developing an adaptive matched filter in the frequency domain to address these challenges effectively.

Leveraging the Short-Time Fourier Transform (STFT) domain, the project will utilize phase shift analysis and magnitude-based separation techniques. By building on the principles of Fast Multichannel Non-Negative Matrix Factorization (FastMNMF), this adaptive filter aims to improve accuracy and robustness in separating sound sources. The proposed implementation will focus on Python-based algorithms, with the goal of optimizing computational efficiency and achieving high-quality separation. The filter's performance will be evaluated using metrics such as Signal-to-Distortion Ratio (SDR), Signal-to-Interference Ratio (SIR), and Signal-to-Artifacts Ratio (SAR).

2. Motivation

Sound source separation has critical applications in fields such as teleconferencing, augmented reality, and audio signal enhancement. While FastMNMF provides efficient magnitude-based separation, its performance in reverberant environments can be limited due to the lack of phase information.

This project seeks to address this gap by integrating phase shift analysis into the STFT domain, enabling improved source differentiation. The focus on Python-based implementation aligns with the project's goal to provide a robust, practical solution that can be applied to real-world audio processing scenarios. By exploring advanced algorithms and leveraging existing libraries, the project aims to contribute to the ongoing development of effective sound separation techniques.

3. Objective

- To develop an adaptive matched filter that accurately separates sound sources in reverberant environments.
- To integrate phase shift analysis with magnitude-based separation in the STFT domain, enhancing separation performance.
- To implement and optimize the filter using Python, focusing on computational efficiency and accuracy.
- To validate the filter's performance using standard metrics, including SDR, SIR, and SAR

4. Methodology

The project will follow a systematic and iterative approach to design, implement, and evaluate the adaptive matched filter. The key steps include:

1. **Literature Review:** Conduct a comprehensive review of existing sound separation techniques, focusing on FastMNMF and related phase-aware methods.
2. **Algorithm Development:** Develop the core algorithm for the adaptive matched filter, integrating phase shift analysis and magnitude-based separation in the STFT domain.
3. **Python Implementation:** Implement the algorithm using Python, leveraging libraries such as NumPy, SciPy, and mir_eval for processing and evaluation.
4. **Testing and Evaluation:**
    - Use standard datasets with multichannel sound sources in reverberant environments.
    - Evaluate performance using SDR, SIR, and SAR metrics.
5. **Iterative Refinement:** Optimize the algorithm based on testing results to improve accuracy and computational efficiency.
6. **Documentation:** Prepare a comprehensive report detailing the methodology, results, and insights gained during the project.

5. Timeline

Tasks	**Target Date**
Literature review	01 Jan – 20 Jan (tentative)
Project Initiation and Algorithm Development	21 Jan - 15 Feb (“)
Python Implementation	16 Feb – 20 Mar (“)
Testing and Evaluation	21 Mar - 01 Apr (“)
Iterations	02 Apr – 25 Apr (“)
Final Report Preparation	01 May – 20 May (“)

6. References

- Weninger, F., et al. "Discriminative NMF and Its Application to Single-Channel Source Separation." *Interspeech 2014.*
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